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Mixture-of-Partitions: Infusing Large Biomedical Knowledge Graphs into BERT

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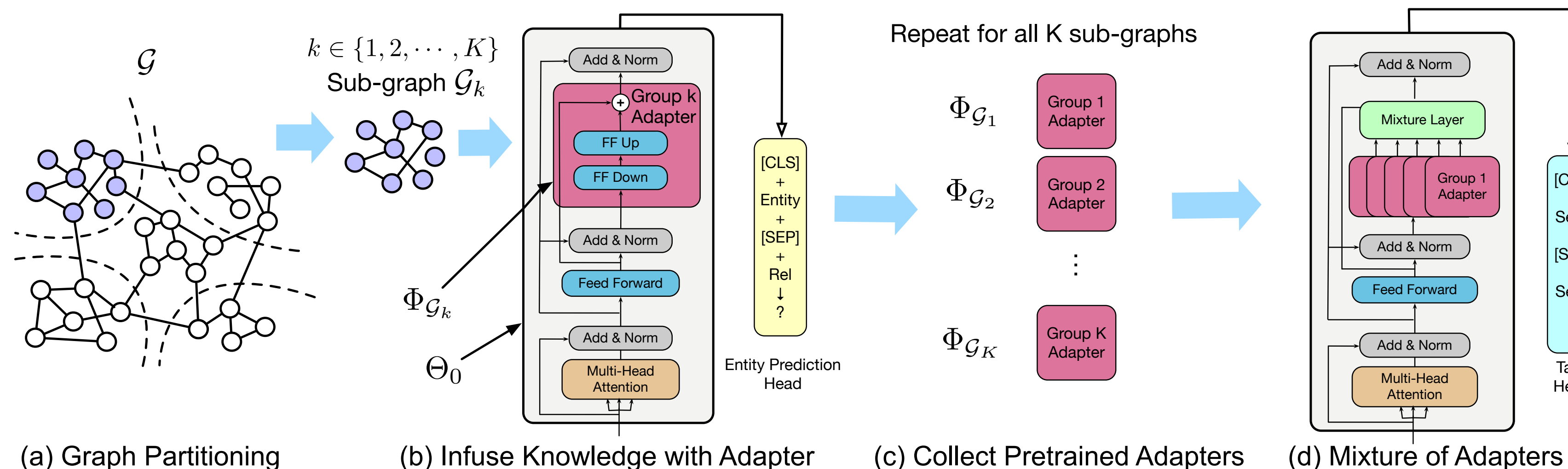
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Highlights

- Use METIS algorithm to partition large biomedical knowledge graphs into smaller subgraphs.
- Use Transformer ADAPTER modules to infuse the factual knowledge to avoid the catastrophic forgetting issue.
- Introduce mixture layers to automatically route useful knowledge from these adapters to down-stream tasks.



(d) **Mixture of Adapters.** AdapterFusion is used to combine knowledge from different adapters by using an attention with the softmax weights:

$$s_{l,k} = \text{Softmax}(\Phi_{l,\mathcal{G}_1} \Phi_{l,\mathcal{G}_2} \cdots \Phi_{l,\mathcal{G}_K}; \Theta_{l,0}), \quad (1)$$

where $s_{l,k}$ is used to mix the adapter outputs to be passed into the next layer, and the final layer L is used to predict a task label y :

$$y = f\left(\sum_{k=1}^K s_{L,k} \Phi_{L,\mathcal{G}_k}\right), \quad (2)$$

where f is the target task prediction head.

Results

Model↓, Dataset→	HoC	PubMedQA	BioASQ7b	BioASQ8b	MedQA	MedNLI
SciBERT	80.52	57.38	75.93	75.72	29.42	81.19
+ MoP (SFull)	81.48↑	54.78	79.14↑	74.74	33.03↑	81.43↑
+ MoP (S20Rel)	81.79↑	54.66	78.50↑	76.25↑	32.77↑	81.20
BioBERT	81.41	60.24	77.50	78.75	30.48	82.42
+ MoP (SFull)	81.47↑	61.82↑	81.29↑	82.04↑	33.55↑	83.44↑
+ MoP (S20Rel)	82.53↑	61.04↑	80.79↑	80.00↑	34.15↑	82.93↑
PubMedBERT	82.25	55.84	87.71	84.54	35.08	84.18
+ MoP (SFull)	82.71↑	61.74↑	88.64↑	86.12↑	36.33↑	84.25↑
+ MoP (S20Rel)	83.26↑	62.84↑	90.64↑	85.39↑	38.02↑	84.70↑
SOTA	82.32	60.24	87.56	90.32	36.70	83.80

Table 1: Performance on various tasks.

Performance on Downstream Tasks. MoP pretrained on the SFull KG improves both the BioBERT and PubMedBERT models for all the tasks, while the SciBERT model can be also improved on 4 out of 6 tasks. The result shows that MoP pretrained with the S20Rel KG achieves new SOTA performances on five evaluated datasets.

Case Study. We can observe that MoP identifies the most related sub-graphs for each example (e.g. Q2 has more weights on [sub-graphs \[1,13,14\]](#), which specialise in ‘tumor’ knowledge). This validates the effectiveness of our MoP in balancing useful knowledge across adapters.

Motivation

- Leveraging factual knowledge to augment pretrained language models is of paramount importance for knowledge-intensive tasks, such as question answering and fact checking.
- Due to the large number of entities in biomedical KGs, computing an exact softmax over all entities is very expensive for training and predicting.
- Different tasks require different domain knowledge, we need to find a way to automatically find useful partial knowledge and combine them to enhance the specific downstream tasks.

Mixture-of-Partitions (MoP)

(a) Graph Partitioning. Three requirements for partitioning: (1) maximize the number of resulting knowledge triples to retain as much factual knowledge as possible; (2) balance nodes over partitions to reduce the overall parameters across different entity prediction heads; (3) efficiency at scale for handling large KGs.

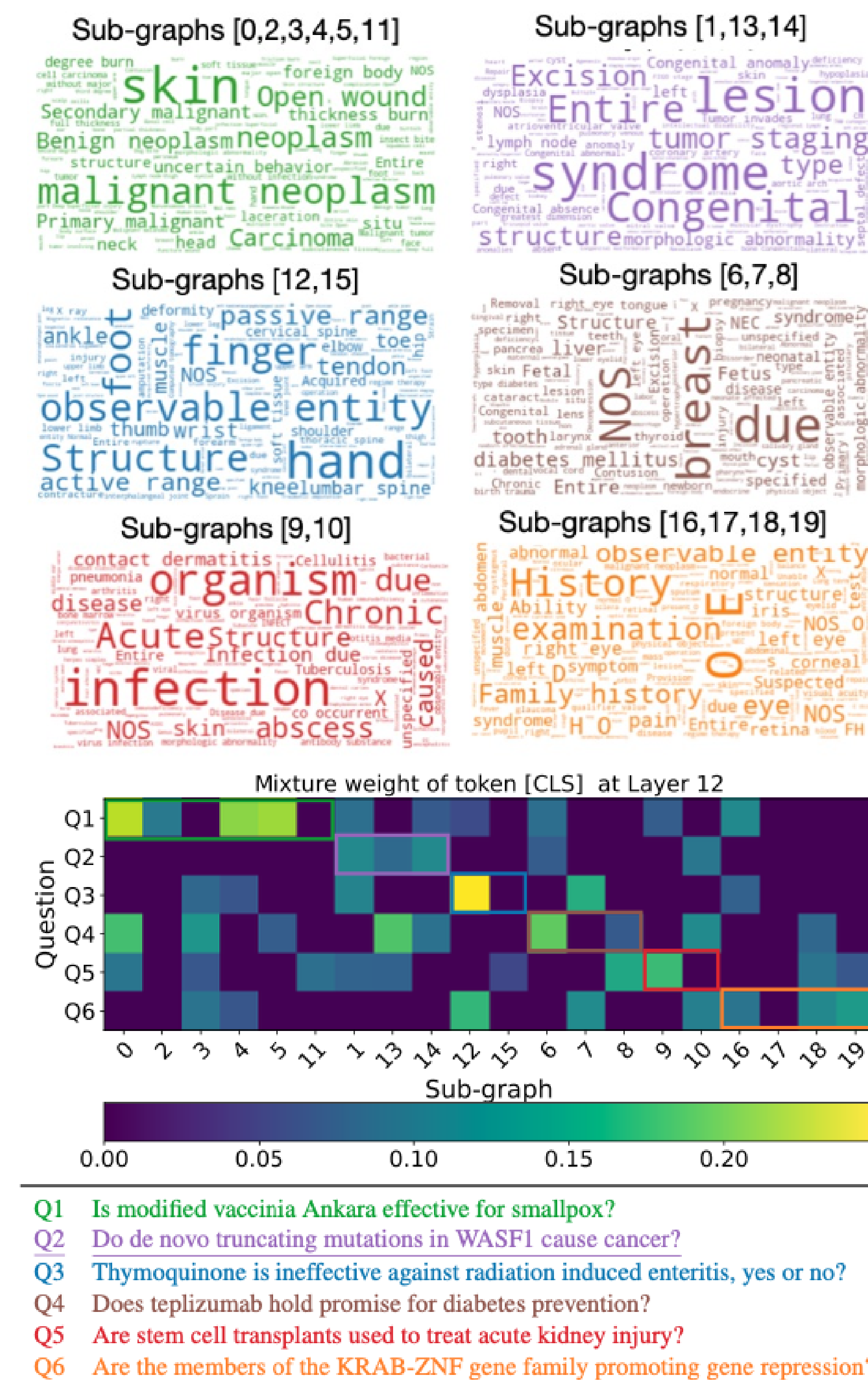
METIS can handle billion-scale graphs by successively coarsening a large graph into smaller graphs, processing them quickly and then projecting the partitions back onto the larger graph.

(b) Knowledge Infusion with ADAPTER. Once the large knowledge graph is partitioned, we use ADAPTER modules to infuse the factual knowledge into a pretrained Transformer model by training an entity prediction objective for each sub-graph. Given a sub-graph $\mathcal{G}_k$, we remove the tail entity name for each triple $(h, r, t) \in \mathcal{G}_k$, and transform the triple into a list of tokens:

“[CLS] h [SEP] r [SEP]”

The sub-graph specific ADAPTER module is trained to predict the tail entity using the representation of the [CLS] token and a cross-entropy loss.

(c) Learn Adapter parameters $\Phi_{\mathcal{G}_k}$ for partitioned each sub-graph $\mathcal{G}_k$.



Takeaways

- Language models pretrained under different partitions of knowledge graph shows a large variance in performance.
- Our MoP can effectively leverage and balance the useful knowledge across those adapters to enhance various downstream tasks.
- Further pruning of the knowledge triples helps task performance by reducing noise, and is a promising direction to explore in future.

Acknowledgements

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